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CSES Problem Set**Shortest Subsequence**TASK [STATISTICS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

You are given a DNA sequence consisting of characters A, C, G, and T.

Your task is to find the shortest DNA sequence that is not a subsequence of the original sequence.

Input

The only input line contains a DNA sequence with n characters.

Output

Print the shortest DNA sequence that is not a subsequence of the original sequence. If there are several solutions, you may print any of them.

Constraints

- $1 \leq n \leq 10^6$

Example

Input:

ACGTACGT

Output:

AAA

CSES Problem Set**Swap Game**TASK [STATISTICS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

You are given a 3×3 grid containing the numbers $1, 2, \dots, 9$. Your task is to perform a sequence of moves so that the grid will look like this:

1	2	3
4	5	6
7	8	9

On each move, you can swap the numbers in any two adjacent squares (horizontally or vertically). What is the minimum number of moves required?

Input

The input has three lines, and each of them has three integers.

Output

Print one integer: the minimum number of moves.

Example

Input:

2	1	3
7	5	9
8	4	6

Output:

4

CSES Problem Set

Multiplication Table

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Find the middle element when the numbers in an $n \times n$ multiplication table are sorted in increasing order. It is assumed that n is odd.

For example, the 3×3 multiplication table is as follows:

1	2	3
2	4	6
3	6	9

The numbers in increasing order are $[1, 2, 2, 3, 3, 4, 6, 6, 9]$, so the answer is 3.

Input

The only input line has an integer n .

Output

Print one integer: the answer to the task.

Constraints

- $1 \leq n < 10^6$

Example

Input:

3

Output:

3

CSES Problem Set

Bubble Sort Rounds I

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Bubble sort is a sorting algorithm that consists of a number of rounds. On each round the algorithm scans the array from left to right and swaps any adjacent elements that are in the wrong order.

Given an array of n integers, calculate the number of bubble sort rounds needed to sort the array.

Input

The first line has an integer n : the array size.

The next line has n integers $x_1, x_2, \dots, x_n$: the array contents.

Output

Print one integer: the number of rounds.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input:

```
5
3 2 4 1 4
```

Output:

```
3
```

Explanation: Bubble sort needs three rounds to sort this array. The array contents after each round are $[2, 3, 1, 4, 4]$, $[2, 1, 3, 4, 4]$, and $[1, 2, 3, 4, 4]$.

CSES Problem Set**Advertisement**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A fence consists of n vertical boards. The width of each board is 1 and their heights may vary.

You want to attach a rectangular advertisement to the fence. What is the maximum area of such an advertisement?

Input

The first input line contains an integer n : the width of the fence.

After this, there are n integers $k_1, k_2, \dots, k_n$: the height of each board.

Output

Print one integer: the maximum area of an advertisement.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq k_i \leq 10^9$

Example

Input:

8
4 1 5 3 3 2 4 1

Output:

10

CSES Problem Set

Special Substrings

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A substring is called *special* if every character that appears in the string appears the same number of times in the substring.

Your task is to count the number of special substrings in a given string.

Input

The only input line has a string of length n . Every character is between a...z.

Output

Print one integer: the number of special substrings.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input:

abccabab

Output:

5

Explanation: The special substrings are abc, cab, abccab, bccaba and ccabab.

CSES Problem Set

Counting LCM Arrays

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given two integers n and k , your task is to count the number of arrays $a_1, a_2, \dots, a_n$ of positive integers where $\text{lcm}(a_i, a_{i+1}) = k$ for all $1 \leq i < n$.

Input

The first line has an integer t : the number of test cases.

The next t lines have two integers n and k : the length of the array and the value of lcm.

Output

Print t integers: the answer to each test case modulo $10^9 + 7$.

Constraints

- $1 \leq t \leq 1000$
- $2 \leq n \leq 10^9$
- $1 \leq k \leq 10^9$

Example

Input:

```
3
3 4
4 6
1337 42
```

Output:

```
11
64
602746233
```

Explanation: The arrays for the first test case are $[1, 4, 1]$, $[1, 4, 2]$, $[1, 4, 4]$, $[2, 4, 1]$, $[2, 4, 2]$, $[2, 4, 4]$, $[4, 1, 4]$, $[4, 2, 4]$, $[4, 4, 1]$, $[4, 4, 2]$ and $[4, 4, 4]$.

CSES Problem Set**Square Subsets**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an array of n integers, count the number of subsets whose elements' product is a square number.

Also count the empty subset with product equal to one.

Input

The first line has an integer n : the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the answer to the problem modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 5000$
- $1 \leq x_i \leq 5000$

Example

Input:

4
2 2 3 6

Output:

4

CSES Problem Set

Stack Weights

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You have n coins, each of which has a distinct weight.

There are two stacks which are initially empty. On each step you move one coin to a stack. You never remove a coin from a stack.

After each move, your task is to determine which stack is heavier (if we can be sure that either stack is heavier).

Input

The first input line has an integer n : the number of coins. The coins are numbered $1, 2, \dots, n$. You know that coin i is always heavier than coin $i - 1$, but you don't know their exact weights.

After this, there are n lines that describe the moves. Each line has two integers c and s : move coin c to stack s ($1 = \text{left}$, $2 = \text{right}$).

Output

After each move, print $<$ if the right stack is heavier, $>$ if the left stack is heavier, and $?$ if we can't know which stack is heavier.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input:

```
3
2 1
3 2
1 1
```

Output:

```
>
<
?
```

Explanation: After the last move, if the coins are $[2, 3, 4]$, the left stack is heavier, but if the coins are $[1, 2, 5]$, the right stack is heavier.

CSES Problem Set

Pyramid Array

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array consisting of n distinct integers. On each move, you can swap any two adjacent values.

You want to transform the array into a *pyramid array*. This means that the final array has to be first increasing and then decreasing. It is also allowed that the final array is only increasing or decreasing.

What is the minimum number of moves needed?

Input

The first input line has an integer n : the size of the array.

The next line has n distinct integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the minimum number of moves.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input:

```
4
2 1 5 3
```

Output:

```
1
```

Explanation: You may swap the first two values which creates a pyramid array $[1, 2, 5, 3]$.

CSES Problem Set

Bit Inversions

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is a bit string consisting of n bits. Then, there are some changes that invert one given bit. Your task is to report, after each change, the length of the longest substring whose each bit is the same.

Input

The first input line has a bit string consisting of n bits. The bits are numbered $1, 2, \dots, n$.

The next line contains an integer m : the number of changes.

The last line contains m integers $x_1, x_2, \dots, x_m$ describing the changes.

Output

After each change, print the length of the longest substring whose each bit is the same.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq x_i \leq n$

Example

Input:

001011

3

3 2 5

Output:

4 2 3

Explanation: The bit string first becomes 000011, then 010011, and finally 010001.

CSES Problem Set

Writing Numbers

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You would like to write a list of positive integers $1, 2, 3, \dots$ using your computer. However, you can press each key $0-9$ at most n times during the process.

What is the last number you can write?

Input

The only input line contains the value of n .

Output

Print the last number you can write.

Constraints

- $1 \leq n \leq 10^{18}$

Example

Input:

5

Output:

12

Explanation: You can write the numbers $1, 2, \dots, 12$. This requires that you press key 1 five times, so you cannot write the number 13.

CSES Problem Set**Maximum Building I**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a map of a forest where some squares are empty and some squares have trees.

What is the maximum area of a rectangular building that can be placed in the forest so that no trees must be cut down?

Input

The first input line contains integers n and m : the size of the forest.

After this, the forest is described. Each square is empty (.) or has trees (*).

Input

Print the maximum area of a rectangular building.

Constraints

- $1 \leq n, m \leq 1000$

Example

Input:

```
4 7
...*.*.
.*.....
.....
.....*
.....*
```

Output:

12

CSES Problem Set**Sorting Methods**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Here are some possible methods using which we can sort the elements of an array in increasing order:

1. At each step, choose two adjacent elements and swap them.
2. At each step, choose any two elements and swap them.
3. At each step, choose any element and move it to another position.
4. At each step, choose any element and move it to the front of the array.

Given a permutation of numbers $1, 2, \dots, n$, calculate the minimum number of steps to sort the array using the above methods.

Input

The first input line contains an integer n .

The second line contains n integers describing the permutation.

Output

Print four numbers: the minimum number of steps using each method.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input:

8
7 8 2 6 5 1 3 4

Output:

20 6 5 6

CSES Problem Set

Cyclic Array

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a cyclic array consisting of n values. Each element has two neighbors; the elements at positions n and 1 are also considered neighbors.

Your task is to divide the array into subarrays so that the sum of each subarray is at most k . What is the minimum number of subarrays?

Input

The first input line contains integers n and k .

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

There is always at least one division (i.e., no value in the array is larger than k).

Output

Print one integer: the minimum number of subarrays.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq k \leq 10^{18}$

Example

Input:

```
8 5
2 2 2 1 3 1 2 1
```

Output:

```
3
```

Explanation: We can create three subarrays: $[2, 2, 1]$, $[3, 1]$, and $[2, 1, 2]$ (remember that the array is cyclic).

CSES Problem Set**List of Sums**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

List A consists of n positive integers, and list B contains the sum of each element pair of list A .

For example, if $A = [1, 2, 3]$, then $B = [3, 4, 5]$, and if $A = [1, 3, 3, 3]$, then $B = [4, 4, 4, 6, 6, 6]$.

Given list B , your task is to reconstruct list A .

Input

The first input line has an integer n : the size of list A .

The next line has $\frac{n(n-1)}{2}$ integers: the contents of list B .

You can assume that there is a list A that corresponds to the input, and each value in A is between $1 \dots k$.

Output

Print n integers: the contents of list A .

You can print the values in any order. If there are more than one solution, you can print any of them.

Constraints

- $3 \leq n \leq 100$
- $1 \leq k \leq 10^9$

Example

Input:

```
4
4 4 4 6 6 6
```

Output:

```
1 3 3 3
```

Explanation: In this case list A can be either $[1, 3, 3, 3]$ or $[2, 2, 2, 4]$ and both solutions are accepted.